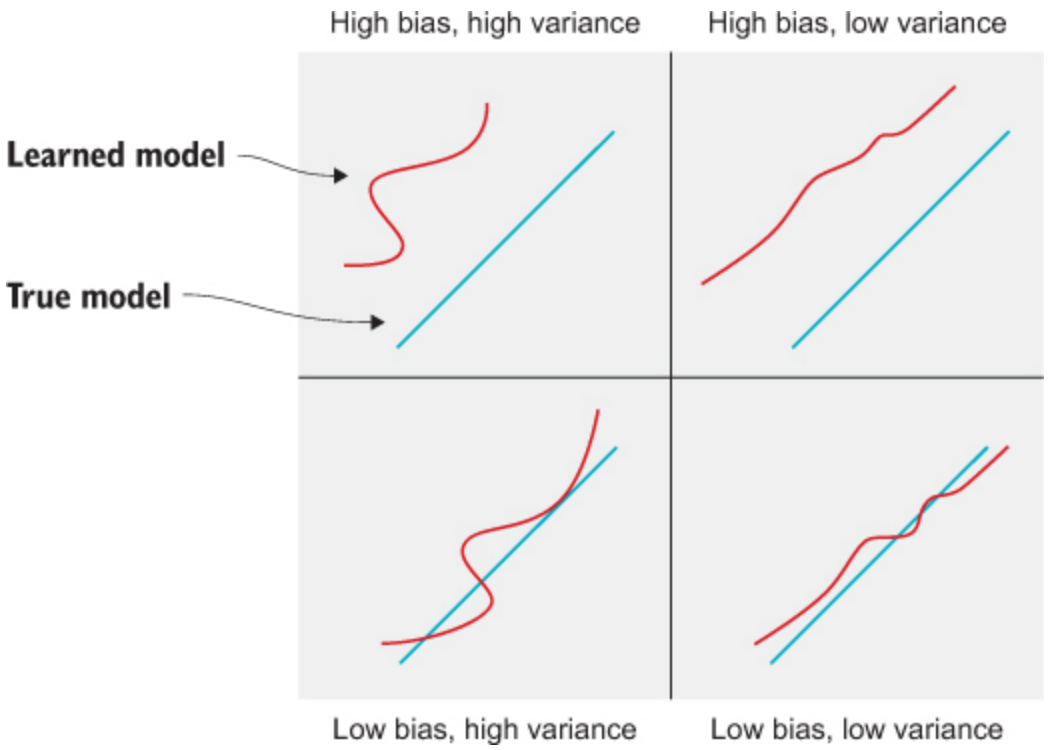
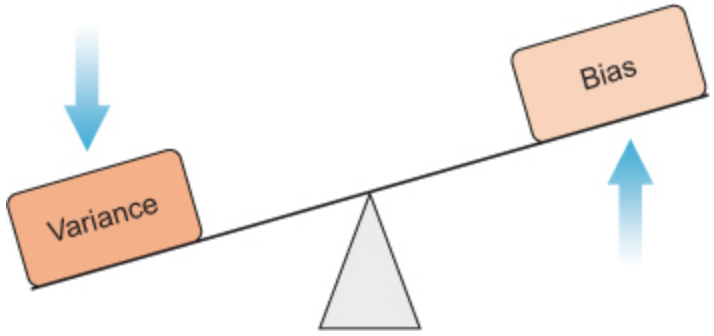


Figure 5.5. The bias-variance tradeoff. Increasing model complexity can reduce bias, but it will increase variance. Reducing variance will increase bias.



We want to combine the potentially high-bias, low-variance value prediction with the potentially low-bias, high-variance policy prediction to get something with moderate bias and variance—something that will work well in the online setting. The role of the critic is hopefully starting to become clear. The actor (the policy network) will take a move, but the critic (the state-value network) will tell the actor how good or bad the action was, rather than only using the potentially sparse raw reward signals from the environment. Thus the critic will be a term in the actor's loss function. The critic, just like with Q-learning, will learn directly from the reward signals